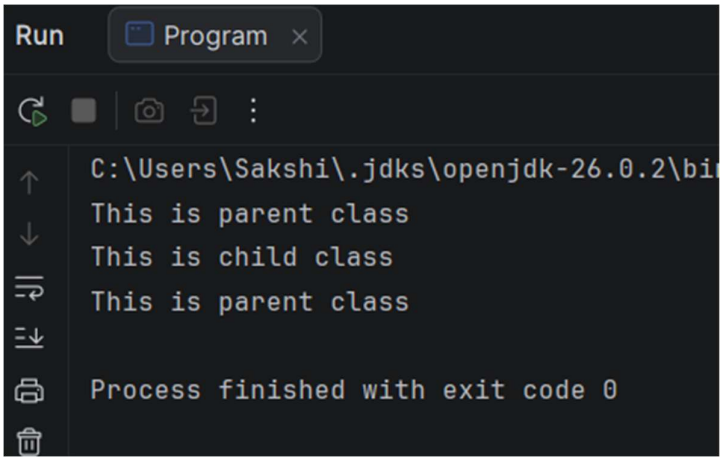
JAVA ASSIGNMENT 2 INHERITANCE

Q1.

CODE:

```
public class Parent {
    void parentMethod(){
        System.out.println("This is parent class");
    }
}
class Child extends Parent {
    void childMethod(){
        System.out.println("This is child class");
    }
}
class Program {
    public  static void main(String[] args){
        Parent p =  new Parent();
        Child c = new Child();
        p.parentMethod();
        c.childMethod();
        c.parentMethod();
    }
}
```

OUTPUT:



Q2.

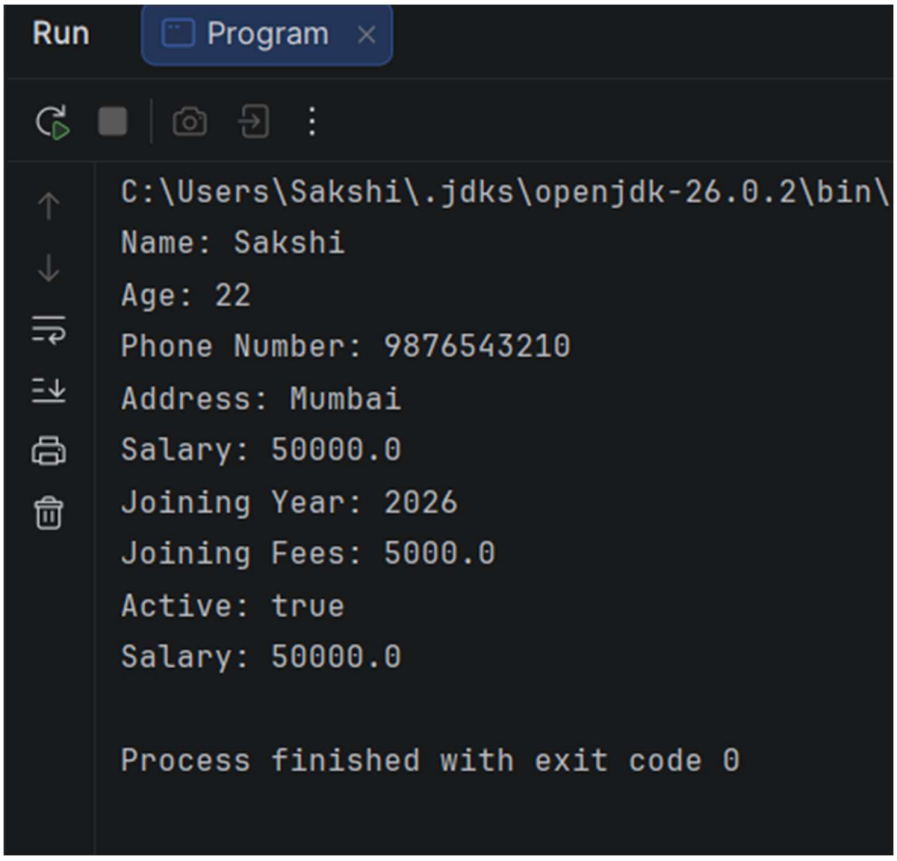
CODE:

```
class Member {
    private String name;
    private int age;
    private String phoneNumber;
    private String address;
    private double salary;
    public void setName(String name) {
        this.name = name;
    }
    public void setAge(int age) {
        this.age = age;
    }
    public void setPhoneNumber(String phoneNumber) {
        this.phoneNumber = phoneNumber;
    }
    public void setAddress(String address) {
        this.address = address;
    }
    public void setSalary(double salary) {
        this.salary = salary;
    }
```

```
    public String getName() {
        return name;
    }
    public int getAge() {
        return age;
    }
    public String getPhoneNumber() {
        return phoneNumber;
    }
    public String getAddress() {
        return address;
    }
    public double getSalary() {
        return salary;
    }
    public void printSalary() {
        System.out.println("Salary: " + salary);
    }
}
class PrimeMembers extends Member {
    private int joiningYear;
    private double joiningFees;
    private boolean isActive;
    public void setJoiningYear(int joiningYear) {
        this.joiningYear = joiningYear;
    }
    public void setJoiningFees(double joiningFees) {
        this.joiningFees = joiningFees;
    }
    public void setActive(boolean active) {
        isActive = active;
    }
    public int getJoiningYear() {
        return joiningYear;
    }
```

```
    public double getJoiningFees() {
        return joiningFees;
    }
    public boolean isActive() {
        return isActive;
    }
    public void display() {
        System.out.println("Name: " + getName());
        System.out.println("Age: " + getAge());
        System.out.println("Phone Number: " + getPhoneNumber());
        System.out.println("Address: " + getAddress());
        System.out.println("Salary: " + getSalary());
        System.out.println("Joining Year: " + joiningYear);
        System.out.println("Joining Fees: " + joiningFees);
        System.out.println("Active: " + isActive);
    }
}
class Program {
    public static void main(String[] args) {
        PrimeMembers p = new PrimeMembers();
        p.setName("Sakshi");
        p.setAge(22);
        p.setPhoneNumber("9876543210");
        p.setAddress("Mumbai");
        p.setSalary(50000);
        p.setJoiningYear(2026);
        p.setJoiningFees(5000);
        p.setActive(true);
        p.display();
        p.printSalary();
    }
}
```

OUTPUT:



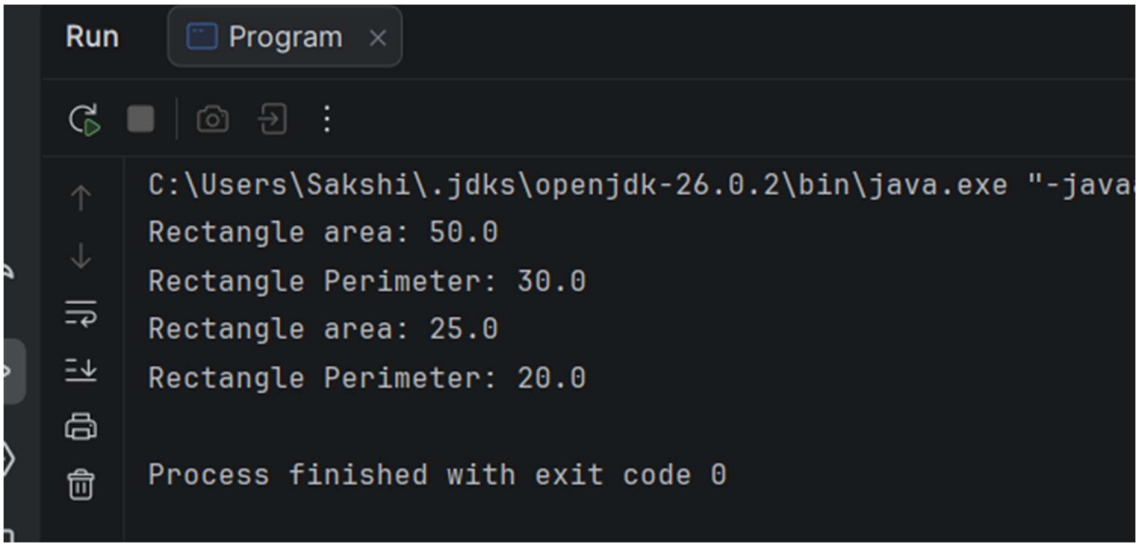
Q3.

CODE:

```
public class Rectangle {
    double length;
    double breadth;
    Rectangle(double l, double b){
        length = l;
        breadth = b;
    }
    void area(){
        System.out.println("Rectangle area: " +(length * breadth));
    }
    void perimeter(){
```

```
        System.out.println("Rectangle Perimeter: " +(2 *(length + breadth)));
    }
}
class Square extends Rectangle{
    Square(double side){
        super(side,side);
    }
}
class Program{
    public static void main(String[] args){
        Rectangle r = new Rectangle(10,5);
        Square s = new Square(5);
        r.area();
        r.perimeter();
        s.area();
        s.perimeter();
    }
}
```

OUTPUT:

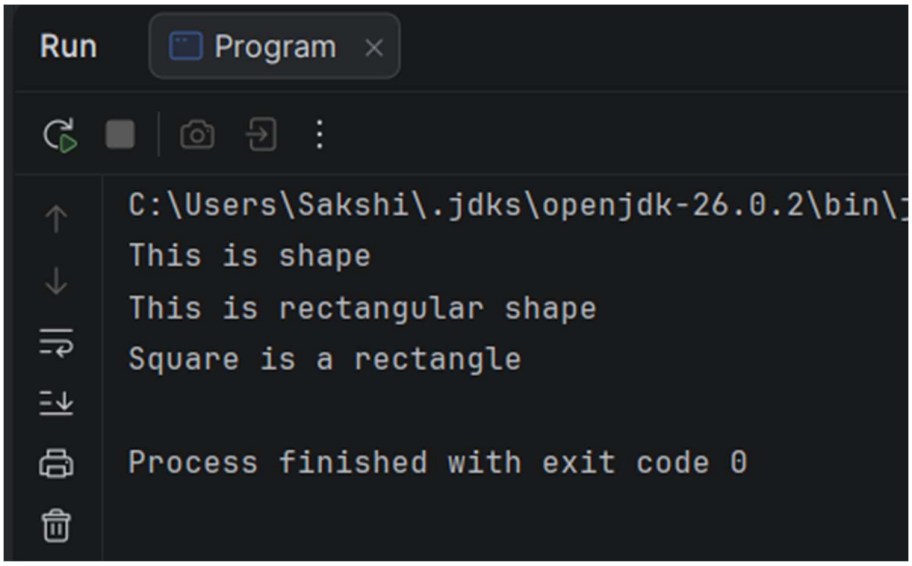


Q4.

CODE:

```
public class Shape {
    void showShape(){
        System.out.println("This is shape");
    }
}
class Rectangle extends Shape {
    void showRectangle() {
        System.out.println("This is rectangular shape");
    }
}
class Circle extends Shape {
    void showCircle() {
        System.out.println("This is circular shape");
    }
}
class Square extends Rectangle {
    void showSquare() {
        System.out.println("Square is a rectangle");
    }
}
 class Program {
    public static void main(String[] args){
        Square s = new Square();
        s.showShape();
        s.showRectangle();
        s.showSquare();
    }
}
```

OUTPUT:


Run
Program
C:\Users\Sakshi\.jdks\openjdk-26.0.2\bin\
This is shape
This is rectangular shape
Square is a rectangle
Process finished with exit code 0